

Capacity versus accessibility

in frozen tanh networks

An accurate readout may exist but require large coefficients or very slow gradient descent. This note develops the bounded-slope result [1] in that framing, proves its sharp bounds, and proposes ordinary GD/Adam audits and a one-step damped Gauss–Newton probe. The new audits are proposals, not reported results.

1 What the theorem measures

Fix $m, W \geq 1$, $\Gamma > 0$, real slopes and biases with $|a_j| \leq \Gamma$, samples $x_i \in [-1, 1]$, and a nonzero target vector. Set

$$A_{i0} = m^{-1/2}, \quad A_{ij} = m^{-1/2} \tanh(a_j x_i + b_j), \quad y_i = f(x_i)/\sqrt{m}, \quad c = M\theta, \quad B = AM. \quad (1)$$

Here c contains the physical output bias and readout weights; W counts tanh neurons. The matrix $M \in \mathbb{R}^{(W+1) \times (W+1)}$ is a fixed invertible readout map, and $L(\theta) = \frac{1}{2} \|B\theta - y\|^2$ is half-MSE. All norms are Euclidean or induced operator norms unless stated. Let P_k project onto sampled polynomials of degree at most k , and $Q_k = I - P_k$. The cutoff k is a polynomial degree, not a width; tails vanish once $P_k = I$.

Capacity versus optimization. Let $\Pi = BB^\top$ project onto the exact feature span. For every θ ,

$$\underbrace{\|B\theta - y\|^2}_{\text{total sampled error}} = \underbrace{\|(I - \Pi)y\|^2}_{\text{unrestricted span error}} + \underbrace{\|B\theta - \Pi y\|^2}_{\text{optimization error inside the span}}. \quad (2)$$

This is Pythagoras: $B\theta - \Pi y$ lies in the span, and $(I - \Pi)y$ is orthogonal to it. Least squares minimizes the first expression to the first term. An invertible M preserves that span but changes its optimization metric, $K = BB^\top$. In particular $B^\top y = B^\top \Pi y$: replacing y by Πy leaves readout gradients unchanged.

For a unit correction q , define its *squared access* $\mu(q) = \|B^\top q\|^2 = q^\top K q$. If $\chi = \|\Pi q\|^2 > 0$ and $\hat{q} = \Pi q / \sqrt{\chi}$, then

$$\mu(q) = \underbrace{\chi}_{\text{span overlap}} \underbrace{\hat{q}^\top K \hat{q}}_{\text{curvature within the span}}. \quad (3)$$

For $\chi = 0$, access is zero. Thus small access can mean missing capacity, small positive curvature, or both. Nearly polynomial features need not have a polynomial span: p and $p + \varepsilon q$ span q , but extracting it costs coefficients of order $1/\varepsilon$. The slope theorem bounds *expressivity per unit coefficient movement*, not unrestricted expressivity.

Theorem 1: necessary tail energy divided by access. Train only θ by unit-rate Euclidean gradient flow, $\dot{\theta} = -B^\top r$, $r = B\theta - y$. For $r_0 \neq 0$ and $0 < \epsilon < 1$, put

$$\delta_k = \frac{\|Q_k r_0\|}{\|r_0\|}, \quad q_k = \frac{Q_k r_0}{\|Q_k r_0\|}, \quad \mu_k = \|B^\top q_k\|^2, \quad T_\epsilon = \inf\{t : \|r(t)\| \leq \epsilon \|r_0\|\}.$$

$$T_\epsilon \geq \underbrace{\frac{\log(1/\epsilon)}{(1-\epsilon)^2}}_{C_\epsilon} \sup_k \underbrace{\frac{[\delta_k - \epsilon]_+^2}{\mu_k}}_{\text{squared readout access}}. \quad (4)$$

necessary tail energy

For $\theta_0 = 0$, $\delta_k = D_k(f) := \|Q_k y\| / \|y\|$, the relative best degree- k polynomial error. The prefactor C_ϵ is optimal for this ratio. Only nonzero tails enter; a positive numerator over zero means $+\infty$, and a zero numerator contributes zero.

The proof is in Section 3. This is a frozen, exact-arithmetic statement in the chosen coefficient metric, not an Adam convergence theorem, an asymptotic numerical floor, or a guarantee that large slopes train quickly.

2 Sharp bounded-slope access: statement and proof

For $\gamma > 0$, define

$$\begin{aligned} \rho_\gamma &= \frac{\pi}{2\gamma} + \sqrt{1 + \frac{\pi^2}{4\gamma^2}}, & \alpha_\gamma^* &= \log \rho_\gamma = \operatorname{arsinh}\left(\frac{\pi}{2\gamma}\right), \\ U_k(\gamma) &= \frac{4\rho_\gamma^{-k}}{\rho_\gamma - 1} \left[\frac{1}{\sqrt{\gamma^2 + \pi^2/4}} + \frac{1}{\pi(k+1)} \right], & e_k(\gamma) &= \min\{\tanh \gamma, U_k(\gamma)\}, \quad e_k(0) = 0. \end{aligned} \quad (5)$$

Theorem 2 (structural bound). For every integer $k \geq 0$, arbitrary real biases, and arbitrary samples,

$$\|Q_k A\|^2 \leq \|Q_k A\|_F^2 \leq \sum_{j=1}^W e_k(|a_j|)^2 \leq W e_k(\Gamma)^2, \quad \mu_k \leq \|Q_k B\|^2 \leq \underbrace{\|M\|^2}_{\mathcal{B}_k} W e_k(\Gamma)^2. \quad (6)$$

The rate and raw width factor are sharp in the sense below; the uniform prefactors are not claimed optimal.

Proof: exact poles and polynomial approximants. Write $t_\ell = \pi(\ell + \frac{1}{2})$, $d_\ell = \sqrt{1 + (t_\ell/\gamma)^2}$, and $\rho_\ell = t_\ell/\gamma + d_\ell$. Logarithmic differentiation of the classical product for $\cosh$ [2] gives

$$\tanh z = \sum_{\ell \geq 0} \left(\frac{1}{z - it_\ell} + \frac{1}{z + it_\ell} \right).$$

Pairs converge locally uniformly away from the poles. For $\phi(x) = \tanh(\gamma x + b)$, the poles are $z_\ell^\pm = (-b \pm it_\ell)/\gamma$, each contributing $-\gamma^{-1}(z_\ell^\pm - x)^{-1}$. For $z \notin [-1, 1]$, choose $s = \sqrt{z^2 - 1}$ and $w = z + s$ with $|w| > 1$. Summing the geometric cosine series and truncating at degree k yields

$$\frac{1}{z - x} = \frac{1}{s} \left(1 + 2 \sum_{n \geq 1} w^{-n} T_n(x) \right), \quad \left\| \frac{1}{z - \cdot} - p_k \right\|_\infty \leq \frac{2|w|^{-k}}{|s|(|w| - 1)}. \quad (7)$$

Put $r = |w| = e^u$ and $v = |\operatorname{Im} z|$. The ellipse parameterization gives $v \leq \sinh u$ and

$$|s|^2(r - 1)^2 = (r - 1)^2 \sinh^2 u + v^2 \left(\frac{2r}{r + 1} \right)^2.$$

Both terms increase with r , whose least admissible value is $r(v) = v + \sqrt{1 + v^2}$. There $|s| = \sqrt{1 + v^2}$. Thus (7) is bounded uniformly in the real part of the pole. Summing conjugate pairs gives

$$\inf_{\deg p \leq k} \|\phi - p\|_\infty \leq R_k(\gamma) := \frac{4}{\gamma} \sum_{\ell \geq 0} \frac{\rho_\ell^{-k}}{d_\ell(\rho_\ell - 1)} \leq U_k(\gamma). \quad (8)$$

The $O(\ell^{-k-2})$ remainders and paired convergence justify the approximant. For the last inequality, the summand decreases with $v = t_\ell/\gamma$: bound the sum by its first term plus its integral with spacing π/γ . Substitute $v = \sinh u$ to obtain

$$\int_{\log \rho_\gamma}^{\infty} \frac{e^{-ku}}{e^u - 1} du = \sum_{n \geq k+1} \frac{\rho_\gamma^{-n}}{n} \leq \frac{\rho_\gamma^{-k}}{(k+1)(\rho_\gamma - 1)},$$

which is exactly (5). The endpoint-midpoint constant approximant has error at most $\tanh \gamma$. Negative slopes reduce by oddness; zero slopes are constant. Projecting each polynomial approximant away gives $\|Q_k A_{:j}\| \leq e_k(|a_j|)$ and the Frobenius bound. Also e_k increases with γ : the first term of U_k is $\frac{8}{\pi} \frac{\rho+1}{\rho^2+1} \rho^{-k}$; both it and the second decrease with $\rho = \rho_\gamma$, which decreases with γ . Finally $\|Q_k A M\| \leq \|Q_k A\| \|M\|$. $\square$

Sharpness. For $\tanh(\Gamma x) = \sum c_n T_n(x)$, the same expansion gives, for odd n ,

$$|c_n| = \frac{4}{\Gamma} \sum_{\ell \geq 0} \frac{\rho_\ell^{-n}}{d_\ell} \geq \frac{4}{\sqrt{\Gamma^2 + \pi^2/4}} \rho_\Gamma^{-n}.$$

Under $dx/(\pi\sqrt{1-x^2})$, the best degree- k error is at least $|c_{n_k}|/\sqrt{2}$, with n_k the next odd degree, so its k th-root limit is ρ_Γ^{-1} . Fine equal-weight Chebyshev grids recover each fixed-degree error, excluding a faster uniform exponent (not a limit on one finite grid). Repeating a nonzero projected column z gives $\|Q_k A\|^2 = W \|z\|^2$, proving the width claim.

3 Coefficient cost, gradient flow, and ordinary GD

A capacity statement with a norm budget. For $q_k^y = Q_k y / \|Q_k y\|$ and $\mu_k^y = \|B^\top q_k^y\|^2$, the best fit within a native-coordinate budget R obeys

$$\mathcal{E}_B(R) := \min_{\|\theta\| \leq R} \frac{\|B\theta - y\|}{\|y\|} \geq \sup_k \left[D_k(f) - \frac{R\sqrt{\mu_k^y}}{\|y\|} \right]_+. \quad (9)$$

Indeed, $\|B\theta - y\| \geq |\langle q_k^y, B\theta - y \rangle| \geq \|Q_k y\| - \sqrt{\mu_k^y} \|\theta\|$. Thus reaching tolerance ϵ requires $R \geq \|y\| \sup_k [D_k - \epsilon]_+ / \sqrt{\mu_k^y}$. This is constrained capacity, not absence from the unrestricted span. For a physical coefficient budget use $M = I$ (or transport the constraint); equal budgets in different native coordinates are not equivalent. Measured coefficient maxima are not Euclidean norms.

Proof of Theorem 1. For any unit q , set $\delta = |\langle q, r_0 \rangle| / \|r_0\|$ and $\mu = \|B^\top q\|^2$. Any iterate reaching relative tolerance ϵ needs movement

$$[\delta - \epsilon]_+ \|r_0\| \leq |\langle q, B[\theta(T) - \theta_0] \rangle| \leq \sqrt{\mu} \|\theta(T) - \theta_0\|. \quad (10)$$

To bound this movement sharply, take a full eigenbasis $Ku_i = \nu_i u_i$ with $\nu_i \geq 0$, and write $r_0 = \sum_i \beta_i u_i$. Solving the linear dynamics gives

$$r(t) = e^{-tK} r_0, \quad \|\theta(t) - \theta_0\|^2 = \sum_{\nu_i > 0} \beta_i^2 \frac{(1 - e^{-t\nu_i})^2}{\nu_i}. \quad (11)$$

At a finite first hitting time $T = T_\epsilon$, continuity gives $\|r(T)\| = \epsilon \|r_0\|$. Set $p_i = \beta_i^2 / \|r_0\|^2$ and $z_i = e^{-2T\nu_i}$. Then $\sum p_i = 1$ and $\sum p_i z_i = \epsilon^2$. Define $F(z) = 2(1 - \sqrt{z})^2 / (-\log z)$, with $F(0) = F(1) = 0$. Equation (11) becomes

$$\frac{\|\theta(T) - \theta_0\|^2}{T \|r_0\|^2} = \sum_i p_i F(z_i).$$

For completeness, F is concave. Put $s = -\frac{1}{2} \log z$ and $g(s) = (1 - e^{-s})^2 / s$. Differentiation yields

$$F''(z) = \frac{g''(s) + 2g'(s)}{4z^2}, \quad g''(s) + 2g'(s) = \frac{2e^{-2s}}{s^3} \underbrace{(s + 1 - e^s)}_{\leq 0} \underbrace{((s - 1)e^s + 1)}_{\geq 0} \leq 0.$$

The second factor vanishes at zero and has derivative $se^s \geq 0$. Jensen therefore gives

$$\|\theta(T) - \theta_0\|^2 \leq T \|r_0\|^2 F(\epsilon^2) = T \|r_0\|^2 \frac{(1 - \epsilon)^2}{\log(1/\epsilon)}. \quad (12)$$

Combine (10)–(12), take $q = q_k$, and maximize over k . Zero eigenvalues are included and require no extra assumption. The residual need not remain polynomial-orthogonal. $\square$

Optimal prefactor. With samples $(-1, 1)$, one feature $\tanh(\gamma x)$, $M = I$, and $r_0 = (-1, 1)^\top / \sqrt{2}$, one has $P_0 r_0 = 0$ and $Kr_0 = \mu r_0$, where $\mu = \tanh^2 \gamma$. Then $T_\epsilon = \log(1/\epsilon) / \mu = C_\epsilon (1 - \epsilon)^2 / \mu$: the prefactor cannot increase. Additional spectral spread can make actual learning much slower.

Discrete GD, not a flow-time surrogate. For $\theta_{n+1} = \theta_n - \eta B^\top r_n$ with $0 < \eta \leq 1/\|B\|^2$, the exact curve and Jensen bound are

$$\|r_n\|^2 = \sum_i \beta_i^2 (1 - \eta \nu_i)^{2n}, \quad N_\epsilon \geq \left\lceil \frac{\log(1/\epsilon)}{-\log(1 - \eta \mu_0)} \right\rceil, \quad \mu_0 = \frac{\|B^\top r_0\|^2}{\|r_0\|^2}, \quad (13)$$

where $0 < \eta \mu_0 < 1$. Convexity of $(1 - \eta \nu)^{2n}$ proves the inequality; equality holds on one eigenspace. For arbitrary tail probes, (10) and the discrete displacement formula also imply the computable necessary condition

$$\max_{0 \leq \nu \leq \|B\|^2} \frac{[1 - (1 - \eta \nu)^n]^2}{\nu} \geq \sup_k \frac{[\delta_k - \epsilon]_+^2}{\mu_k}, \quad (14)$$

with value zero at $\nu = 0$. These are GD statements; ordinary Adam is an empirical comparison, not covered by the time bounds.

4 Gamma scaling and a damped-GN accessibility probe

Constant slopes versus width-scaled slopes. For a shared positive slope, $\Gamma = \gamma$; with heterogeneous slopes it is a maximum, not a median. At fixed Γ_0 , the raw squared-access envelope has the form $C(\Gamma_0)W \exp(-2\alpha_{\Gamma_0}^* k)$. If a relevant cutoff is $k = \kappa W$, this is exponentially small in width. For $\Gamma = cW$, instead,

$$k\alpha_{\Gamma}^* = \kappa W \operatorname{arsinh}\left(\frac{\pi}{2cW}\right) \longrightarrow \frac{\pi\kappa}{2c}. \quad (15)$$

Thus width-scaled slopes remove this exponential suppression from the bound; they do not guarantee large actual access or fast optimization. The full prefactor matters: for example, if $k + 1 \geq \Gamma$, (5) gives $U_k(\Gamma) \leq \frac{8}{\pi}(1 + 1/\pi)e^{-k\alpha_{\Gamma}^*}$. Readout maps contribute $\|M\|^2$; polynomially growing maps do not cancel an exponential obstruction.

Why might k scale with width? If the *necessary* target tail satisfies two-sided $D_k \asymp e^{-\beta k}$ and the requested accuracy is $\epsilon_W \asymp e^{-sW}$, choose a cutoff with $D_k \geq 2\epsilon_W$ just below the required degree. Then $k = (s/\beta)W + O(1)$, and the energy/access ratio has exponential factor $e^{2(\alpha_{\Gamma}^* - \beta)k}$. It grows for fixed Γ when $\alpha_{\Gamma}^* > \beta$. This is conditional: fixed-frequency entire targets can have faster tails, and fixed tolerance need not require growing degree. Measure D_k and maximize the certificate rather than assume $k = W$. In particular, these results do not prove failure for every $\Gamma = o(W)$, nor necessity of linear slopes for every target.

Proposition 3 (one-step damped Gauss–Newton). Because the residual is affine in θ , its exact Hessian and GN matrix both equal $B^\top B$. For $\zeta > 0$, solve one fixed-damping subproblem in the chosen readout metric:

$$\Delta\theta_\zeta = \arg \min_d \left\{ \frac{1}{2} \|r + Bd\|^2 + \frac{\zeta}{2} \|d\|^2 \right\}, \quad r_\zeta^+ = \zeta(K + \zeta I)^{-1}r. \quad (16)$$

For any unit correction q with access $\mu(q) = q^\top Kq$, define the remaining fraction $\mathcal{R}(q; \zeta) = \|\zeta(K + \zeta I)^{-1}q\|$. Then

$$\boxed{\mathcal{R}(q; \zeta) \geq \frac{\zeta}{\mu(q) + \zeta}, \quad q \in \operatorname{range}(Q_k) \implies \mathcal{R}(q; \zeta) \geq \frac{\zeta}{\mathcal{B}_k + \zeta}.} \quad (17)$$

Proof. The normal equations give $\Delta\theta_\zeta = -(B^\top B + \zeta I)^{-1}B^\top r$, implying (16). For $q = \sum_i \omega_i u_i$ with $\sum \omega_i^2 = 1$,

$$\mathcal{R}(q; \zeta)^2 = \sum_i \omega_i^2 \frac{\zeta^2}{(v_i + \zeta)^2} \geq \frac{\zeta^2}{(\sum_i \omega_i^2 v_i + \zeta)^2}.$$

This is Jensen for the convex function $x \mapsto \zeta^2/(x + \zeta)^2$. Equality holds on a single eigenspace, so the bound is sharp given only $\mu(q)$. The structural bound proves the second inequality. $\square$

Damping is a soft *curvature* cutoff, not a polynomial-degree or Fourier cutoff: a mode of curvature v retains amplitude $\zeta/(v + \zeta)$. Reducing a unit correction to fraction ϵ requires $\zeta \leq \epsilon\mu(q)/(1 - \epsilon)$. With relative damping $\rho = \zeta/\|B\|^2$ and normalized access $\hat{\mu} = \mu/\|B\|^2$, the bound is $\mathcal{R} \geq \rho/(\hat{\mu} + \rho)$. Recovery at larger relative damping indicates easier access relative to the fastest curvature.

Separating missing capacity from weak positive modes. As $\zeta \downarrow 0$, $\mathcal{R}(q; \zeta) \rightarrow \|(I - \Pi)q\|$. The limiting remainder is the probe's span error; the approach to that limit measures its weak positive modes. An isolated polynomial tail need not lie in the span even if the full target does. Test both common tails and the full target, and use Πy as an attainable-target diagnostic. An exact undamped step merely reproduces the least-squares residual. Sweep one step from the same residual at each damping, without adaptive damping, repeated convergence, or a line search.

Interpretation. Gamma can change both the span and optimization within it. The theorem detects small output change per unit coefficient movement. With a norm budget it restricts capacity; with GD dynamics it forces a time cost. The damping probe measures the curvature scale at which the same necessary corrections become recoverable. No single worst condition number separates these effects.

5 Frozen-geometry audit: proposed experiments

Question and primary outcome. How many digits does increasing gamma buy for *ordinary GD* and *ordinary Adam*, after controlling readout coordinates and initialization? Plot independent-grid relative L_2 error versus gamma at fixed budgets. Least-squares refits diagnose available approximation, not an all-or-nothing success criterion. All settings below are proposed; PR #2 supplies the maps and initialization distinctions, not results for this gamma sweep [3].

Matched geometry. Start with the branch's core $N = 512$, $h = 2/N$, halo $H = \lceil \sqrt{N} \rceil = 23$, and $W = N + 1 + 2H = 559$ neurons. Use centers $-1 + jh$, $j = -H, \dots, N + H$, and freeze $a_j = \gamma$, $b_j = -\gamma t_j$ for $\gamma \in \{1, 2, 4, 8, 16, 32, 50, 64\}$. The reference $\lambda = \gamma h = 0.25$ gives $\gamma = 64$; report N, W, h, λ . Train on $16N + 1$ endpoint-inclusive points; evaluate on separate 32,768 midpoints. Start with $\sqrt{2} \sin(2\pi x)$, $\sqrt{5}x^2$, and $[\sin(2\pi x) + 0.1 \sin(20\pi x)]/\sqrt{0.505}$ [3]; add Runge as a new target. Quadratic is a zero-tail control for $k \geq 2$. Follow up at $N = 256, 1024$, comparing constant gamma with fixed λ before accuracy floors.

Readout maps and starts are separate ablations. Keep reference allowances α fixed at $\lambda_{\text{ref}} = 0.25$ across gamma. Let $D = \text{diag}(\sqrt{\alpha})$, $s_j = \sum_{\ell \leq j} \alpha_\ell$ over neurons only, and L be the anchored difference map. Test the branch maps [3]:

Coordinates	Physical hidden readout w	Physical output bias
Raw	$w = \theta_w$	$b = \theta_b$
Collective / individual	$w = D_w \theta_w / w = D_w^2 \theta_w$	$\sqrt{\alpha_0} \theta_b / \alpha_0 \theta_b$
Collective / individual neighboring	$w = L \text{diag}(\sqrt{s}) \theta_w / w = L \text{diag}(s) \theta_w$	Same respective bias scales

Neighboring keeps the final tanh anchor; scale cumulative coefficients *before* differencing. On selected cells, test unscaled bias/ordinary halo scales to expose those penalties. Cross the maps with physical starts $w_0 = 0$, $w_{0,j} = \sqrt{\alpha_j} \sigma_X \xi_j$, and $w_{0,j} = \alpha_j \sigma_X \xi_j$, where $\sigma_X = \sqrt{2/(W+1)}$ and $\xi_j \sim N(0, 1)$; $b_0 = 0$. Reuse draws and encode identical physical weights across maps, rather than redraw neighboring coordinates. The last start is a *new independent alpha-Xavier ablation*: the branch's individual-scale comparison re-encoded the collective start. Its signed-envelope start is optional, not the same Gaussian initialization [3].

Training and theorem measurements. Run full-batch FP64 GD with no momentum and $\eta = c/\|B\|^2$ ($c = 1$ primary, $1/2$ control). Run ordinary Adam [4] with $(\beta_1, \beta_2) = (0.9, 0.999)$, no weight decay, and prescribed constant/decayed schedules; calibrate learning rate and epsilon with equal pilot budgets at small and large gamma. No Armijo, fallback direction, or in-training solve. Compare $10^4, 5 \times 10^4, 2 \times 10^5$ updates, final error and late-window mean/median/excursions, and paired seeds for nonzero starts. Log native and physical norms; do not impose the same numerical norm cap in different metrics.

Build P_k by orthonormal polynomial QR. Measure $D_k, \mu_k, \|Q_k B\|^2, \mathcal{B}_k$, and their curvature-normalized versions up to $k = 256$. Use common unit polynomial probes of degrees 16, 32, 64, 128 and necessary target tails; use r_0 , not y , for nonzero-start certificates. Compare (4), (14), and the exact GD curve (13) with training; flow time is not an iteration count. Report (2), norm-budget fits (9), and projected-target diagnostics to separate span error from learning cost.

Damping audit and decision rule. Sweep one GN step at $\rho = \zeta/\|B\|^2 = 10^0, 10^{-2}, \dots, 10^{-24}$, refining resolved recovery thresholds. Keep each probe fixed across gamma/maps, and test full targets as well as tails. Plot remaining fraction versus damping with (17) and span-limit checks; also record native/physical step norms. The desired evidence is lower GD/Adam error at larger gamma *and* recovery of the same necessary corrections at larger relative damping, surviving the readout controls. A tiny necessary tail, weak certificate, or disappearing gamma effect limits the explanation; distance to the least-squares floor alone does not measure significance.

Numerical controls. Factor B , not a rounded Gram matrix; use SVD/augmented QR for damping. Recompute difficult cells from feature evaluation onward at 80 then 120 digits and vary cutoffs. Discarded singular modes are not exact nullspace; unresolved tails/damping thresholds are not zero. Check map identities, raw/neighbor evaluation agreement, and the single-mode equalities above. Report budget-limited plateaus, not proved floors. No LSMR or optimizer bake-off is part of this audit.

[1] Continuous MLPs project. *Sharp readout access bounds and frozen-geometry audits*. Supplied draft, 19 September 2026.

[2] NIST. *DLMF*, Eq. 4.36.2: infinite product for cosh.

[3] samlayton99/precision-mlps, PR #2 @ b71a03396990. Pinned sources: core and initialization; difference maps; parameter-scale report.

[4] D. P. Kingma and J. Ba. *Adam: A Method for Stochastic Optimization*. ICLR 2015. arXiv:1412.6980.